

# Abhinandan

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## EXPERIENCE

### Founding Software Engineer

Sept 2025 – Present

*Browser*

*Remote, San Francisco*

- Built Browser's Chrome MV3 recorder + CDP-native browser automation agent having AX/DOM element capture with cross-iframe support, obstruction checks, and real mouse/key/upload execution.
- Implemented a smart streaming ReAct loop across FastAPI + extension with SSE tool execution, multi-tab orchestration, safe parallelism, abort/continue, and audit logs.
- Cut cost per automation run 3x (compact traces, context compression, prompt caching, Haiku/Sonnet/GPT-5 routing) across 700K+ monitored production LLM calls.
- Shipped a zero-LLM replay engine: recordings run as variable-driven tool-call templates, with a stateful AI fallback that resumes mid-run on failure.
- Shipped self-healing docs that auto-repair on UI drift. Haiku → Sonnet diff triage, LLM-free replay of intact steps, and a CDP agent that fixes only what changed.

### Founding Software Engineer

Jan 2025 – Jul 2025

*Cynos Nexus*

*Noida, India*

- Built the AI knowledge base service using FastAPI, LangChain, and vector retrieval pipelines, powering customer-facing search workflows.
- Developed document-ingestion pipelines using Google Cloud Vision, XLSX processing, and BullMQ workers, enabling automated extraction of customer data from spreadsheets and scanned records.
- Implemented backend infrastructure for retrieval-augmented AI workflows, including document ingestion, structured data processing, asynchronous workers, Redis-backed queues, and production API serving.
- Automated containerized deployment and CI/CD workflows for AI services using Docker, GitHub Actions, Nginx reverse proxying, and cloud infrastructure.

### Contract Software Engineer

Sept 2024 – Dec 2024

*Etkin.ai*

*Remote, Turkiye*

- Built a LangChain + pgvector retrieval system; raised recall@5 from 0.71 → 0.82 on a 300-query labeled eval set.
- Developed scalable data-ingestion pipelines using bulk CSV processing and Celery workers, reducing processing time by 40%.
- Engineered a production PDF generation system transforming structured AI outputs and dynamic JSON reports into enterprise-grade documents.
- Automated deployment of AI services using Docker, GitHub Actions, and AWS EC2, establishing reliable CI/CD workflows for production environments.

### Open Source Contributor

June 2024 – Aug 2024

*HeroUI*

*Y Combinator S24 startup*

- Received a **personal offer from the CEO** to join HeroUI (prev. NextUI) after making **open-source contributions**.
- Resolved 10+ bugs & delivered 7+ feature enhancements in core components including Calendar, Table and Pagination.

### SWE Intern

Jan 2024 – Apr 2024

*SkilledUp*

*Noida*

## OPEN SOURCE PROJECTS

**AgentFlow-Pro** (*Research*) | *PyTorch, TRL, DAPO, PRM, LoRA* | [github.com/awesome-pro/agentflow-pro](https://github.com/awesome-pro/agentflow-pro)

- Rebuilt and extended the ICLR 2026 *AgentFlow* architecture as a local Qwen3-8B Planner → Executor → Verifier → Memory loop with grammar-constrained planning, Tavily search, and sandboxed Python/SymPy tools.
- Trained the planner with DAPO + a learned Qwen3-0.6B Process Reward Model, replacing the paper's GRPO setup with a custom dynamic-sampling stage and full A40 training pipeline.
- Built a leakage-free, quantization-matched evaluation flow and measured +5.0 pts on GPQA-Diamond, improving from 40.0% to 45.0%.

**smartmemo** | *PyTorch, sentence-transformers, FAISS, SQLite* | [github.com/awesome-pro/smartmemo](https://github.com/awesome-pro/smartmemo)

- Built a semantic memory/cache for LLM agents where embedding retrieval proposes candidates and a learned pair-equivalence classifier decides whether memory reuse is safe.
- Trained classifier-v2 on 16,576 pairs across 9 domains, improving precision by +30 points at equal recall versus cosine similarity, with feedback export, bad-hit detection, gated retraining, and WAL-backed SQLite hardening.

**agenteval** | *AsyncIO, OpenAI/Anthropic/LangChain adapters* | [github.com/awesome-pro/agenteval](https://github.com/awesome-pro/agenteval)

- Agent evaluation toolkit replacing exact-match asserts with repeated-run pass-rate tests; traces tool calls/timing/steps, supports collect-then-raise behavioral assertions, OpenAI/Anthropic/LangChain adapters, and Typer CLI JSON reports for CI gates.

**guardloop** | *OpenAI/Anthropic SDKs, OpenTelemetry, LangGraph* | [github.com/awesome-pro/guardloop](https://github.com/awesome-pro/guardloop)

- Production guardrail runtime for async agents: pre-flight Decimal cost/token/time/tool-call budgets, per-tool circuit breakers, verifier feedback retry loop, OpenTelemetry GenAI spans, and no-rewrite LangGraph/OpenAI Agents SDK adapters.

TECHNICAL SKILLS

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**ML Engineering:** PyTorch, Hugging Face, sentence-transformers, embeddings, RAG, FAISS, Pinecone, Qdrant  
**Agents & Frameworks:** LangGraph, LangChain, LlamaIndex, CrewAI  
**LLM Engineering:** Fine-tuning (LoRA, QLoRA, PEFT), quantization (AWQ, GPTQ, GGUF), inference engines (vLLM, SGLang, Ollama, TGI)  
**Evals & Reliability:** OpenTelemetry, Langfuse, Arize Phoenix, LangSmith  
**Backend & Infrastructure:** FastAPI, Node.js, WebSockets, AWS, GCP, Docker, Redis, PostgreSQL  
**Languages:** Python, TypeScript, C++, Bash

AWARDS & RECOGNITION

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**International Youth Math Challenge Gold Honour | Top 1% TypeScript Engineer Globally (Algora) | Amazon ML Summer School 2025**

EDUCATION

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**Dr. APJ Abdul Kalam Technical University** B.Tech, Computer Science | 2026